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Pricing Strategies of Supermarkets in Uruguay

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Note for the english Version

This version was translated from the original dissertation for dissemination purposes. The original version in Spanish can be accessed in [Ideas](#).

ABSTRACT

This work aims to contribute to the discussion on price-setting behavior among retailers in Uruguay. Using a dataset compiled by the General Directorate of Commerce (from January 2011 to December 2019), I focus on analyzing the strategies of price-setting at the chain level.

The main results show that, on average, the chains change 26 % of their prices each month, with positive price variations being more frequent than negative ones. Results indicate an inverse relationship between the frequency and size of price adjustments.

Furthermore, the analysis at the store level shows that those belonging to the same chain follow a similar price-setting strategy.

The synchronization analysis found heterogeneity between the different chains. The FK index ranges from 0.52 to 0.75.

Survival analysis and hazard functions are upward-sloping or hump-shaped in all chains, consistent with state-dependent pricing.

Keywords:

price synchronization, retail, Hazard rate, Pricing Strategies, Price position.

Capítulo 1

Introduction

During the first half of the last century, Latin American countries experienced a generalized and accelerated increase in prices, reaching hyperinflation levels¹ in some cases. Uruguay was not exempt from this issue and, like its regional peers, began implementing stabilization plans in the 1990s.

Uruguay successfully transitioned from high inflation² to low inflation³ in 2001, according to [Aboal et al. \(2003\)](#). In 2019, inflation was moderate⁴, ranking as the third highest inflation rate among South American countries, based on World Bank data. In this context, studying inflationary mechanisms is of interest for understanding the potential effects of policy interventions.

Uruguayan literature offers several contributions focusing on the recent evolution of inflation and monetary policy [Aboal et al. \(2003\)](#), [Valentini \(2007\)](#), wage-to-price pass-through [Dominguez et al. \(2012\)](#), [Borraz et al. \(2020\)](#), exchange rate pass-through [Gianelli \(2011\)](#), and pricing strategies [Borraz et al. \(2012\)](#), among others. This paper belongs to the latter group and seeks to empirically characterize pricing strategies for different products in Uruguayan supermarkets.

In particular, this paper seeks to address the following questions: Are pricing strategies homogeneous within supermarkets belonging to the same chain? What is the degree of synchronization of these strategies?

Answering these questions is important because it helps to understand the

¹International standards generally define hyperinflation as inflation exceeding 50 % per month or 100 % per year.

²Inflation exceeding 10 % per month.

³Inflation below 10 % per month.

⁴Inflation in Uruguay in 2019 was 8.79 % accumulated over the last 12 months, according to data from the National Institute of Statistics (INE).

impact of monetary policy on real variables.

Using monthly data collected by the Directorate General of Commerce (DGC) on retail prices from twelve supermarket chains during the period January 2011 to December 2019, the results indicate heterogeneity in pricing strategies across chains and some degree of homogeneity within supermarkets belonging to the same chain. An inverse relationship is found between the frequency and magnitude of price changes, with price increases occurring more frequently than decreases.

Applying the price-change synchronization index derived by [Fisher and Konieczny \(2000\)](#), the results reveal varying levels of synchronization. On a scale from 0 to 1, where 0 indicates that prices change at completely different points in time and 1 indicates that all prices change simultaneously, the minimum value obtained is 0.52 and the maximum is 0.75 across different chains. Finally, the hazard function and survival curve were estimated, showing that supermarket chains appear to align with 'State-Dependent Pricing' models.

The document is organized into five chapters. Chapter 2 presents the theoretical and empirical background related to price-setting in different countries. Chapter 3 describes the methods employed in this study. Chapter 4 briefly introduces the data and reports the results obtained. Finally, Chapter 5 provides concluding remarks.

Capítulo 2

Background

In the macroeconomic literature, there are two types of models that explain the aggregate behavior of prices. The first model is **Time-Dependent Pricing (TDP)**, which states that “the probability that a firm changes its prices is a function that depends solely on time” ([Walsh \(2017\)](#), p.277).

The second type, known as **State-Dependent Pricing (SDP)**, states that firms adjust their prices according to the state of the economy and that price adjustment entails an associated cost. For example, the “menu cost” theory illustrates this situation by referring to the cost incurred when a restaurant changes its prices. When calculating potential future benefits, the restaurant must consider the cost of printing new menus and informing the service staff about the changes.

Within the class of **TDP** models, there are two formalizations. The first type assumes that price changes occur in a staggered manner (as described in [Taylor \(1979\)](#)). The second formalization assumes that firms adjust prices simultaneously (sticky pricing), as described in [Calvo \(1983\)](#).

In general terms, in the [Calvo \(1983\)](#) model, nominal shocks are not persistent, whereas in the [Taylor \(1979\)](#) model, they have a lasting effect on real variables.

Both theoretical models assume that firms adjust their prices according to a predetermined frequency, without responding to external shocks.

On the other hand, among the **State-Dependent Pricing (SDP)** models is the work of [Caplin and Leahy \(1991\)](#), one of the first models of “economic dynamics with state-dependent prices in which monetary shocks have effects on output” ([Caplin and Leahy \(1991\)](#), p.684), assuming fixed costs of price

adjustment. The authors conclude that monetary shocks affect either the level of output or the price level, but not both. In a general equilibrium setting, the real effect is permanent.

The theoretical model presented in [Golosov and Lucas Jr \(2007\)](#) simulates idiosyncratic shocks that affect productivity to examine their effects on inflation. They compare the model’s estimated predictions with US inflation data from 1988 to 1997. The study concludes that (i) idiosyncratic shocks account for most price adjustments and (ii) the number of firms changing their prices increases as inflation rises.

Another model within the **SDP** framework is the model of [Dotsey et al. \(1999\)](#) (hereafter DWK),¹ which illustrates the differences between TDP and SDP models. The distribution of the optimal price depends on current and future prices as well as real marginal costs, with the behavior of the price distribution being similar across both models. The difference lies in the assumption that, under TDP theory, the fraction of firms adjusting their prices is fixed over time, making prices more rigid (*sticky*). In contrast, under SDP theory, the number of firms adjusting their prices is random, allowing more firms to change prices in response to a shock. In the DWK model, a monetary shock has a real effect on money in the short run, but not in the long run.

In summary, under SDP theory, firms choose when to change prices subject to “menu costs,” which causes adjustments to be delayed and larger than the optimal level. In contrast, in TDP models, price changes are exogenous and occur at a predetermined frequency, that is, every n periods² or randomly³.

At the empirical level, various studies have attempted to characterize price-setting mechanisms. [Klenow and Kryvtsov \(2008\)](#) conducts an analysis using micro-data from the U.S. Consumer Price Index (CPI) between 1988 and 2004, finding that price changes are frequent (every 4 to 7 months), but the magnitude and probability of price changes are unrelated to time.

At the retail level, the seminal work of [Nakamura and Steinsson \(2008\)](#) uses CPI micro-data from the United States to compare price behavior with menu cost theory. The authors observe that various idiosyncratic shocks affect prices and that the frequency of price changes is highly seasonal.

[Klenow and Malin \(2010\)](#) provides a review of the theoretical literatu-

¹It assumes that the costs of changing prices are fixed for all firms and that the decision to adjust prices occurs randomly across firms.

²[Taylor \(1979\)](#)

³[Calvo \(1983\)](#)

re, comparing empirical results obtained from different data sources (surveys, scanner data, web scraping) in the United States and European countries. These empirical studies focus on the frequency, size, and dynamics of price changes, finding that the frequency of price changes varies considerably across countries. Eurozone countries exhibit the lowest frequency of price changes, while developing countries with high inflation (Brazil, Chile, Mexico, Sierra Leone, Slovakia) show the highest frequency.

[Gopinath and Itskhoki \(2010\)](#) use micro-data on prices of imported goods to the United States and find heterogeneity in the frequency of price adjustments, identifying two distinct behaviors in response to exchange rate shocks. Firms that are further from their optimal price adjust more frequently, whereas those closer to their optimal price adjust less often.

[Wulfsberg \(2016\)](#) analyze micro-data from Norway’s Consumer Price Index over a thirty-year period to identify the behavior of price frequency and magnitude across different inflation levels. The study concludes that at high inflation levels, prices adjust more frequently but with smaller magnitudes. In contrast, when inflation is low, price adjustments occur less frequently but with larger magnitudes.

[Berger and Vavra \(2018\)](#) use micro-data underlying the U.S. consumer, producer, and import price indices to examine the dynamics of price changes. The study finds that the frequency of price adjustments is countercyclical.

Currently, the greater availability of information through the internet has enabled a series of studies based on data collection from various websites.

[Cavallo \(2009\)](#) compares prices in Argentina, Brazil, Chile, and Colombia during the period from October 2007 to October 2008. These economies are characterized by moderate inflation and represent a shift from previous studies that focused on low-inflation contexts. The authors find that the magnitude of price changes is unrelated to the level of inflation. Brazil and Chile show evidence consistent with the predictions of SDP models. The slope of the price-change hazard curve differs from that found by [Nakamura and Steinsson \(2008\)](#), as these countries exhibit higher inflation than the United States and Europe, and inflationary shocks cause greater deviations over time. Additionally, the study observes a high degree of price synchronization in Argentina, Brazil, Chile, and Colombia.

[Chaumont et al. \(2011\)](#) study price-setting behavior in 288 supermarkets and 22,000 goods in Chile over a 24-month period during an inflationary episo-

de. They find a high degree of flexibility and heterogeneity across store chains. Along the same lines for Chile, [Lira et al. \(2008\)](#) investigates the relationship between market structure and price-setting.

Focusing on the consequences of the expansion of two supermarket chains, they find that market concentration is associated with price increases, despite the presence of economies of scale.

[Nunes \(2010\)](#) analyzes price adjustment behavior for 410 goods in Brazil over a 160-month period, finding that the frequency of price changes and inflation are positively related, while the synchronization of price changes is negatively correlated. [Resende et al. \(2014\)](#) examines price synchronization in the Brazilian tire market, tracking the evolution of prices for four different brands and seven types of tires over a four-year period. They find similar synchronization patterns across brands.

Although empirical evidence for the region has increased in recent years, the number of studies focusing on pricing strategies and synchronization in Uruguay remains limited. [Borraz et al. \(2012\)](#) analyzes price dynamics from April 2007 to December 2010 using daily data on 117 products from 303 stores nationwide. They find a high frequency of price changes, which are non-seasonal and concentrated on the first day of the month. Additionally, they observe a high degree of synchronization across chains.

This study aims to contribute further to the line of research in Uruguay, focusing on price-change strategies and synchronization at the chain level, and examining the existence of heterogeneity or similarities among stores belonging to the same chain.

Capítulo 3

Methodology

This section details the tools used to study the frequency and magnitude of price changes (Section 3.1.1), price duration (Section 3.1.2), synchronization of price changes (Section 3.1.3), and long-term price duration (Section 3.2).

3.1. Indicators of Price Change and Synchronization

The main indicators used to measure synchronization, variation, and price duration are described next.

3.1.1. Frequency and Price Variation

A key aspect to examine in pricing strategies is the frequency of price changes.

Following the empirical literature of Nakamura and Steinsson (2008), a dichotomous indicator (0/1) is constructed to indicate whether a good changed its price between two consecutive periods. The frequency is calculated as the average of this variable across the entire sample.

$$\mathbb{1}_{i,j,t} = \begin{cases} 1 & \text{if } p_{ijt} \neq p_{ijt-1} \\ 0 & \text{if } p_{ijt} = p_{ijt-1} \end{cases} \quad (3.1)$$

Where j denotes the store, i the product, and t the months.

Since a price change can result from two situations—price increases or decreases—it is possible to construct indicators that incorporate the direction

of the change.

$$\mathbb{1}_{i,j,t}^p = \begin{cases} 1 & \text{if } p_{i,j,t} > p_{i,j,t-1} \\ 0 & \text{if } p_{i,j,t} \leq p_{i,j,t-1} \end{cases} \quad (3.2)$$

$$\mathbb{1}_{i,j,t}^n = \begin{cases} 1 & \text{if } p_{i,j,t} < p_{i,j,t-1} \\ 0 & \text{if } p_{i,j,t} \geq p_{i,j,t-1} \end{cases} \quad (3.3)$$

Again, the average of $\mathbb{1}_{i,j,t}^p$ and $\mathbb{1}_{i,j,t}^n$ can be interpreted as the frequency of positive and negative price changes, respectively.

In the context of an empirical study of price changes using monthly data, the estimated frequency reflects the probability that a good changes its price between two consecutive months.

The works of [Nakamura and Steinsson \(2008\)](#), [Bils and Klenow \(2004\)](#), [Gopinath and Rigobon \(2008\)](#), and [Cavallo \(2009\)](#) additionally propose using the implicit duration of price changes, given by the inverse of the frequency, which indicates how many periods are required to observe a change in all prices.

The second aspect to examine is the magnitude of price changes. For each store (j), the variation in the price of good (i) at time (t) relative to the previous period ($t - 1$) is calculated as:

$$\Delta_{i,j,t} = \left(\frac{p_{i,j,t}}{p_{i,j,t-1}} - 1 \right) \times 100 \quad (3.4)$$

The indicator can be interpreted as the percentage variation between two consecutive periods. In this study, monthly reports will be used.

It is essential to note that when calculating averages, positive and negative movements can offset each other. Therefore, it is additionally proposed to analyze the size of variations by distinguishing whether they correspond to increases or decreases.

Since this work aims to provide evidence on price-change strategies at the chain level, the previous indicators can be aggregated following [Chaumont et al. \(2011\)](#).

Let A be the chain of interest; its frequency of price changes is given by:

$$F_A = \frac{1}{N_{jA}} \sum_{j \in A} \left(\frac{1}{N_{Ij}} \left(\sum_{i \in I_j} \left(\frac{1}{T_{ij} - 1} \sum_{t \in T_{i,j}} \mathbb{1}_{i,j,t} \right) \right) \right) \quad (3.5)$$

where N_{jA} is the number of stores in chain A , N_{Ij} the total number of

goods sold in store j , I_j the set of products sold, and T_{ij} the total number of periods during which good i was sold in store j .

Similarly, it is possible to obtain an indicator of the magnitude of price changes for each chain:

$$\Delta_A = \frac{1}{N_{jA}} \sum_{j \in A} \left(\frac{1}{N_{Ij}} \left(\sum_{i \in I_j} \left(\frac{\sum_{t \in T_{i,j}} \mathbb{1}_{i,j,t} \times \Delta_{i,j,t}}{\sum_{t \in T_{i,j}} \mathbb{1}_{i,j,t}} \right) \right) \right) \quad (3.6)$$

These two indicators provide a weighted measure for each chain of the frequency of price changes and their magnitude. Additionally, it is also possible to obtain aggregations at the store level.

As with the individual study of prices, it is possible to differentiate between positive and negative frequencies, as well as to analyze the size of price changes according to the direction of the change.

3.1.2. Duration Analysis

To determine the average time during which a price remains unchanged, we followed the strategy used by [Cavallo \(2009\)](#), applying Survival Analysis techniques to identify the time between price adjustments in firms¹.

This technique can be used to study the time elapsed between price adjustments for each good.

Following the notation of [Baumgartner et al. \(2005\)](#), the function $S(t)$ represents the probability that the price “survives” unchanged until period t . In turn, n_j is the number of prices that have not yet changed and are therefore at risk of changing at t_j , while d_j is the number of observed price changes in period t_j .

$$\hat{S}(t) = \prod_{j \leq t} \left[\frac{n_j - d_j}{n_j} \right] \quad (3.7)$$

In the context of price-setting, the cumulative function $S(t)$ indicates the percentage of prices that last for t periods or more. For this reason, a negative slope is expected in this function, particularly a steeper decline when prices have shorter durations.

¹For this study, we first used the `Survival` package library, which includes the non-parametric estimator of the survival function proposed by Kaplan-Meier ([Therneau et al. \(1994\)](#)).

Additionally, to assess whether supermarket strategies are closer to SDP or TDP models, the Hazard function was estimated.

[Nakamura and Steinsson \(2008\)](#) use these estimates to address the question of whether prices that have recently changed are more likely to change again, or whether, on the contrary, the probability increases the longer they remain unchanged.

The Hazard function at t indicates the probability that a good changes its price after t periods, conditional on it not having been modified previously. If the probability of change increases the longer the price remains constant, it is expected that the Hazard function will have a positive slope, indicating that price-setting follows an SDP model ([Nakamura and Steinsson \(2008\)](#), p.1457).

Conversely, observing a flat slope would indicate that price-setting is aligned with the TDP model, meaning it does not depend on time, as in Calvo's model ([Nakamura and Steinsson \(2008\)](#), [Calvo \(1983\)](#)).

Formally, price evolution can be considered as a Markov process² given by (J_n, T_n) , where J_n denotes successive state changes, T_n the entry times into a state, and $S_n = T_n + T_{n-1}$ the duration in the previous state.

In the context of price changes, two possible states are defined for goods: E if the price remains unchanged, and C if it changes between two consecutive periods.

The transition probabilities from state E to C are given by $p_{EC} = P(J_{n+1} = C \mid J_n = E)$, which indicates the probability of moving to state C , given that the previous state was E .

To illustrate this process of state transition, consider the evolution of a hypothetical good that starts in state E at $t = 1$ with a price p . If at $t = 2$ the price remains p , then the good stays in state E , with no change of state. If at $t = 3$ the price becomes $p^* \neq p$, then the good enters state C with an associated entry time of 3 and a previous duration $S = 2$, indicating how many periods it remained in the previous state before changing.

In particular, the interest lies in the hazard rate, given by the probability of a price change conditional on the price having remained unchanged.

Formally, the goal is to estimate the probability of transition from state E to C between period d and Δd , given that the process has remained in state E for d periods, that is:

²For the estimation of Hazard functions, the Semi-Markov package was used [Listwon and Saint-Pierre \(2015\)](#).

$$\lambda_{CE}(d) = \lim_{\Delta d \rightarrow 0} \frac{P(J_{n+1} = C, d < S_{n+1} \leq d + \Delta d \mid J_n = E, S_{n+1} > d)}{\Delta d} \quad (3.8)$$

Assuming that the duration times follow a known distribution (for example, Weibull), it is possible to estimate the parameters of its distribution and the associated Hazard function using maximum likelihood.

3.1.3. Price Synchronization

To provide a measure of price synchronization, we calculated the synchronization indicator proposed by [Fisher and Konieczny \(2000\)](#) and reviewed by [Dias et al. \(2005\)](#).

Assuming the existence of a homogeneous good sold by N firms over T time periods, let p denote the proportion of firms that change prices in period t . The FK indicator can be calculated as:

$$FK = \sqrt{\frac{1}{T} \frac{\sum_{t=1}^T (p_t - \bar{p})^2}{\bar{p}(1 - \bar{p})}} \quad (3.9)$$

This indicator takes values between 0 and 1. It can be interpreted as the proportion of firms in the economy that are synchronized. If the indicator equals 0, price changes across firms are uniformly staggered. If the index equals 1, perfect synchronization occurs, meaning all firms change prices at the same time.

It is important to note that the above indicator assumes the existence of a single homogeneous good and N firms. To obtain an aggregate indicator by chain, we follow [Chaumont et al. \(2011\)](#), calculating synchronization for each good (i) sold in stores (j) belonging to chain A , and taking the simple average of all goods as the measure of synchronization for the chain:

$$FK_A = \frac{1}{N_{IA}} \sum_{i \in I_A} FK_{i,j} \quad (3.10)$$

Where N_{IA} refers to the total number of goods sold by chain A , and $FK_{i,j}$ represents the level of synchronization among stores belonging to the chain with respect to the sale price of good i .

3.2. Price Positioning

Hartveld (2016) propose studying price positioning strategy as the relationship between the price at which good i is sold in store j relative to the market price observed at time t .

$$PP_{i,j,t} = \frac{P_{i,j,t}}{MP_{i,t}} \quad (3.11)$$

Where P represents the price in store j for good i at time t , and MP is the median market price. The indicator PP will take values greater than 1 if the price is above the median and less than 1 if it is below.

This approach has the advantage of allowing aggregations that do not depend on the nominal values of goods. Following the logic of aggregation by chains or stores, it is possible to determine whether a store or chain, on average, sets prices above or below those observed in the market.

Let A be the chain of interest; its price positioning index is given by:

$$PP_A = \left(\frac{1}{N_{jA}} \sum_{j \in A} \left(\frac{1}{N_{Ij}} \sum_{i \in I_j} \frac{1}{T_{ij}} \sum_{t \in T} PP_{i,j,t} \right) \right) \quad (3.12)$$

where N_{jA} is the number of stores in chain A , N_{Ij} the total number of goods sold in store j , and T_{ij} the total number of periods during which good i was sold in store j .

As with other aggregations, it is also possible to compute price positioning at the store level.

Capítulo 4

Empirical Analysis

4.1. Data Used

For this study, price data collected by the Directorate General of Commerce (DGC) were used. To obtain a homogeneous and continuous set of goods, 154 items that have been consistently reported over time were selected. The detailed list of products is presented in Table 1 in the Appendix. The database contains products reported from January 2011 to December 2019. Although the original database includes daily prices, we opted to work with the monthly mode of prices to avoid the effect of temporary discounts or promotional price reductions, focusing instead on systematic patterns.

The data collection and systematization process was previously described in [Borraz et al. \(2012\)](#) (p. 7). The database is updated as follows: “Two days after the end of the month, each supermarket uploads its price information to the DGC. Then a process called price verification begins. This process starts with the calculation of the average price of each item in the basket. Any price that is 40 % higher or lower than the average is flagged. The supermarket is then contacted to verify whether the reported price is correct. If no response is received from the supermarket, or if the supermarket confirms the reported price, it is published on the website as reported. If the supermarket corrects the price—which is exceptional—the database is updated and the corrected price is published online.”

Each row in the database represents the reported selling price of a good in a specific store belonging to a particular chain or group of establishments. Although goods differ in terms of presentation, size, or brand, it is possible

to create additional groupings. For example, “Río de la Plata corn oil, 900 milliliters” can be grouped under the category “food” and the product group “sunflower oil.”

As in [Borraz et al. \(2012\)](#), we followed the criterion of including in the sample only those chains with at least five stores and an average of more than three cash registers per store. Additionally, pharmacy chains were excluded from the database, and reports from 2012 onward were used to create a more homogeneous product base.

This process resulted in a database of 154 goods, grouped into 51 products and 4 categories. A total of 267 stores across the country were considered, grouped into 12 chains.

Table 4.1 presents the number of observations for the period from January 2011 to December 2019, according to the four main categories: (i) food, (ii) alcoholic beverages, (iii) non-alcoholic beverages, and (iv) hygiene and cleaning products (hereafter referred to as “cleaning”). The second column represents the percentage within the total sample, while the last column refers to the weight of these aggregates in the Consumer Price Index¹.

Tabla 4.1: Number of observations by category.

Category	Number of observations	Percentage of total	Weight in CPI
Food	1,834,970	70 %	23 %
Alcoholic beverages	139,468	5 %	1 %
Non-alcoholic beverages	147,861	6 %	3 %
Cleaning	500,841	19 %	3 %
Total	2,623,140	100 %	30 %

Source: Author’s calculations based on DGC and INE data

¹Based on the breakdown of weights by ‘Divisions, Groups, Classes, Families, and Products’ of the Consumer Price Index for the entire country, as reported by the National Institute of Statistics (INE).

4.2. Results

4.2.1. Price Variation

Table 4.2 presents the results of the analysis in Section 3.1.1. The column “Frequency” shows the estimates of the weighted average frequency of price changes for the different chains, as defined in Equation (3.5). The column “Implicit Duration” displays the implicit duration², which is the inverse of the frequency and indicates the average time in months between price changes. Finally, the column “Delta”³ reports the average value of price increases, calculated according to Equation (3.6).

On average, chains change 26 % of prices each month, which corresponds to an implicit duration of 3.99 months. When prices change, the average variation is 3.89 %.

Tabla 4.2: Frequency and Size of Price Changes by Chain

Chain	Frequency	Implicit Duration	Delta
Devoto	0.24	4.14	3.82
Disco	0.24	4.11	3.98
El Clon	0.23	4.35	4.32
El Dorado	0.32	3.15	2.89
Frigo	0.17	5.94	4.78
Macromercado Mayorista	0.31	3.21	3.06
Micro Macro	0.35	2.88	2.33
Multi Ahorro	0.27	3.74	3.28
Red Market	0.20	4.93	5.10
Ta - Ta	0.30	3.33	3.40
Tienda Inglesa	0.24	4.17	5.62
Ubesur	0.26	3.90	4.12
Average	0.26	3.99	3.89

Source: Author’s calculations based on DGC data.

When analyzing these results at the chain level, it is observed that chains that tend to make more frequent price changes are those that implement changes of smaller magnitude. In other words, there is an inverse relationship between the frequency of price changes and their size (Figure 4.1). For example, the chain Frigo makes price changes of greater magnitude but less frequently.

²The Annex (Table 4) presents the results of price duration by category, and the Annex (Table 5) shows price duration by product.

³The Annex (Table 2) presents the results of the size of price changes by category, and the Annex (Table 3) shows the size of price changes by product.

Figura 4.1: Average Size and Frequency of Price Changes by Chain

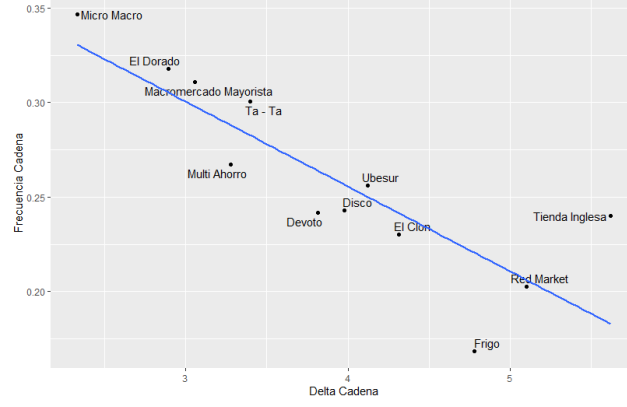
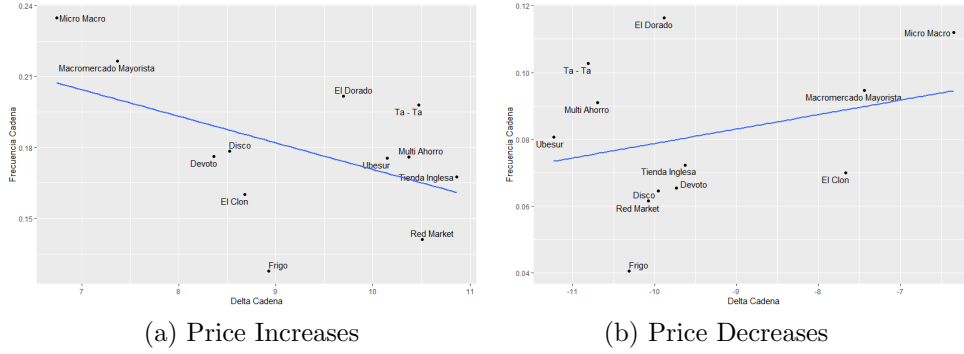


Figura 4.2: Average Size and Frequency of Price Changes by Chain and Sign of Change



Price changes were analyzed by separating increases from decreases. Table 4.3 performs the same analysis, decomposing by the sign of the change. It is noteworthy that the frequency of positive changes is higher than that of negative changes, indicating that prices tend to rise more often than they fall. Additionally, this decomposition allows us to break down the frequency calculated in Table 4.2. For example, the chain Micro Macro, which exhibited the highest frequency of price changes (35 %), can be approximately decomposed into $\frac{2}{3}$ price increases and $\frac{1}{3}$ price reductions. The heterogeneity in frequencies and the size of changes may be due to different pricing strategies. However, it is worth highlighting that chains that increase prices more frequently tend to do so in smaller magnitudes, while chains that implement larger price reductions do so less frequently (Figure 4.2).

Tabla 4.3: Frequency and Size of Price Changes by Chain and Sign of Change

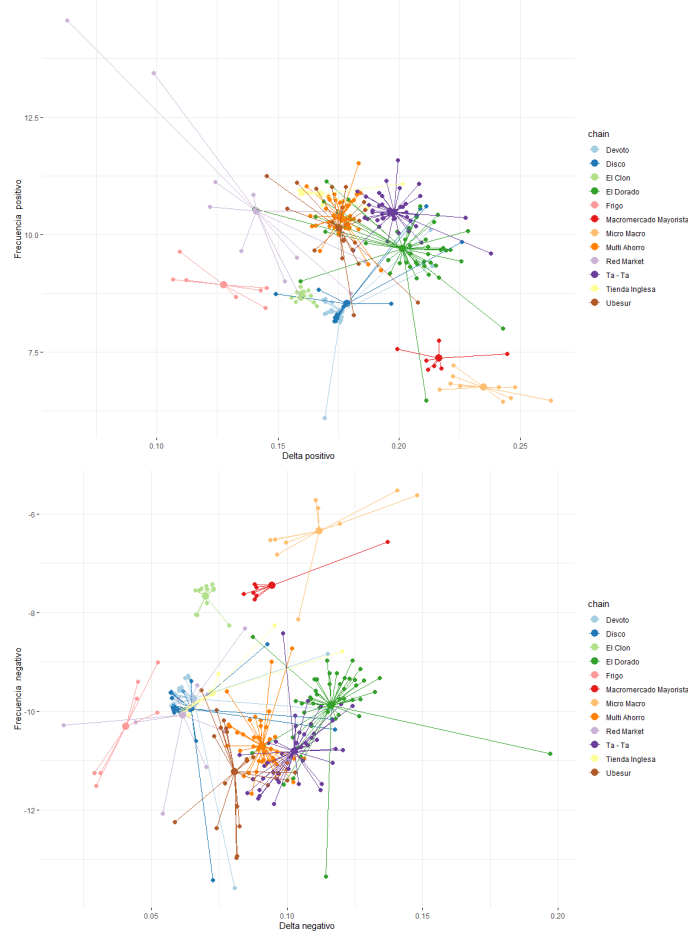
Chain	Frequency		Delta	
	Positive	Negative	Positive	Negative
Devoto	0.18	0.07	8.36	-9.74
Disco	0.18	0.06	8.52	-9.95
El Clon	0.16	0.07	8.68	-7.67
El Dorado	0.20	0.12	9.70	-9.88
Frigo	0.13	0.04	8.92	-10.31
Macromercado Mayorista	0.22	0.09	7.37	-7.44
Micro Macro	0.23	0.11	6.74	-6.35
Multi Ahorro	0.18	0.09	10.37	-10.70
Red Market	0.14	0.06	10.51	-10.08
Ta - Ta	0.20	0.10	10.47	-10.81
Tienda Inglesa	0.17	0.07	10.86	-9.63
Ubesur	0.18	0.08	10.15	-11.23

Source: Author's calculations based on DGC data.

To investigate whether these pricing strategies are driven by chain-level policies, the same analysis was performed by aggregating at the store level.⁴ The graph provides evidence that these strategies indeed appear to be defined at the chain level. Observations belonging to different chains tend to cluster together, and there is a low level of mixing among stores from different chains (Figure 4.3).

⁴The Annex (Figure 1) presents the results of average frequency and average price changes by supermarket, additionally including information on chain affiliation and whether the store is located in Montevideo or in the interior of the country.

Figura 4.3: Average Size and Frequency of Price Changes



(a) Monthly Positive Changes (b) Monthly Negative Changes

To provide more detailed information, the decomposition of price changes at the chain level (Figure 4.3) is presented in Table 4.4. Column 1 of Table 4.4 shows the number of stores within each chain. Column 2 reports the average delta for price decreases. Column 3 shows the average frequency of price decreases. Column 4 presents the standard deviation of the delta for price decreases. Column 5 shows the standard deviation of the frequency of price decreases. Column 6 provides the standardized distance of stores relative to the chain average in both dimensions, that is, a measure of how clustered the chains are. Column 7 reports the average delta for price increases. Column 8 shows the average frequency of price increases. Column 9 presents the standard deviation of the delta for price increases. Column 10 shows the standard deviation of the frequency of price increases. Finally, Column 11 reports the

standardized distance of stores in both dimensions for price increases.

Results show that Frigo has, on average, a higher frequency of price decreases, while Red Market exhibits the highest frequency of price increases. Regarding the standardized distance, which indicates the degree of clustering among the different chains, El Clon shows the lowest values for both price decreases and price increases. Conversely, stores belonging to Ubesur are more dispersed in terms of price decreases, and Red Market is more dispersed with respect to price increases.

Tabla 4.4: Statistics on Average Size and Frequency of Price Changes

Chain	n	Price Decreases					Price Increases				
		Average		Std. Dev.		Std. Dist.	Average		Std. Dev.		Std. Dist.
		Delta	Freq	Delta	Freq	Dist.	Delta	Freq	Delta	Freq	Dist.
	1	2	3	4	5	6	7	8	9	10	11
Devoto	24	-9.7375	0.0655	0.8495	0.0161	0.8317	8.3608	0.1762	0.6440	0.0117	0.6305
Disco	28	-9.9546	0.0646	0.7599	0.0165	0.7464	8.5239	0.1784	0.6695	0.0148	0.6576
El Clon	14	-7.6702	0.0700	0.2597	0.0035	0.2503	8.6802	0.1602	0.1149	0.0030	0.1108
El Dorado	44	-9.8826	0.1164	0.8255	0.0163	0.8162	9.6966	0.2016	0.7700	0.0206	0.7615
Frigo	7	-10.3126	0.0406	1.0101	0.0104	0.9353	8.9248	0.1277	0.3764	0.0176	0.3489
Macromercado Mayorista	7	-7.4421	0.0946	0.3991	0.0188	0.3699	7.3665	0.2165	0.2331	0.0138	0.2162
Micro Macro	10	-6.3486	0.1120	0.7724	0.0190	0.7330	6.7436	0.2348	0.2353	0.0152	0.2237
Multi Ahorro	48	-10.6968	0.0911	0.5681	0.0066	0.5622	10.3725	0.1761	0.3903	0.0060	0.3862
Red Market	12	-10.0791	0.0615	0.9139	0.0173	0.8752	10.5064	0.1412	1.8404	0.0357	1.7623
Ta - Ta	43	-10.8128	0.1027	0.6770	0.0104	0.6692	10.4735	0.1980	0.3832	0.0106	0.3789
Tienda Inglesa	10	-9.6293	0.0724	0.6445	0.0201	0.6117	10.8632	0.1675	0.1529	0.0147	0.1457
Ubesur	20	-11.2294	0.0807	0.9876	0.0102	0.9627	10.1454	0.1754	0.8240	0.0134	0.8032

Fuente: calculado por la autora en base a datos de la DGC.

4.2.2. Duration Analysis

To understand pricing strategies over time, Survival and Hazard functions were calculated as a function of time, as detailed in Section 3.1.2.

Table 4.5 presents duration as the weighted average number of months that a good remains without a price change by chain. This measure offers an alternative approach to comparing chain strategies and price rigidities.

Tabla 4.5: Price Duration by Chain

Chain	Average Duration
Devoto	3.96
Disco	3.98
El Clon	3.47
El Dorado	2.47
Frigo	5.30
Macromercado Mayorista	2.79
Micro Macro	2.24
Multi Ahorro	2.72
Red Market	4.54
Ta - Ta	2.39
Tienda Inglesa	3.18
Ubesur	3.27

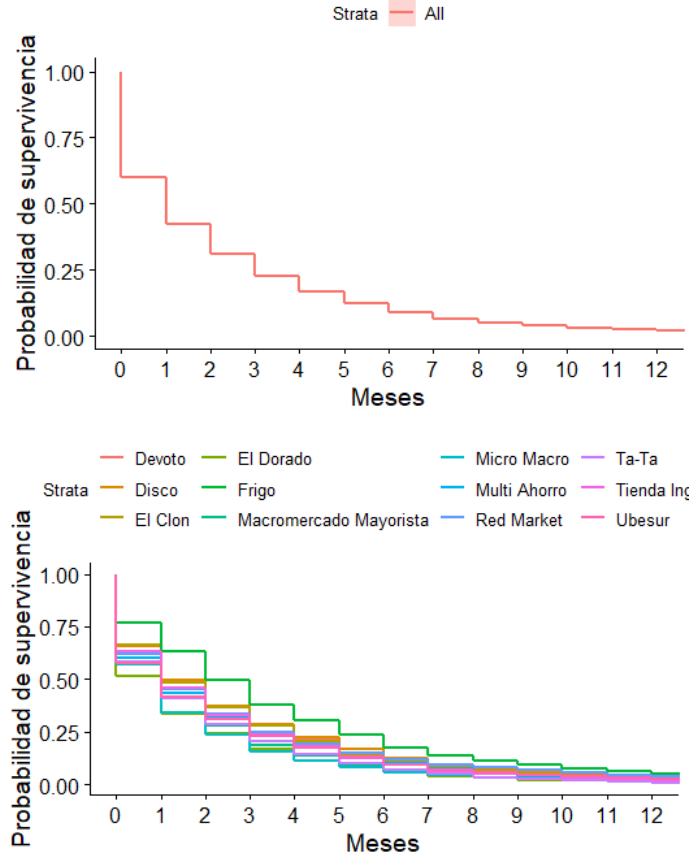
Source: Author's calculations based on DGC data.

As shown in Table 4.5, there is heterogeneity among chains, with Micro Macro being the chain with the lowest price rigidity, while Frigo exhibits the highest. These results are consistent with the previous section, where chains that changed prices more frequently implemented changes of smaller magnitude.

Figure 4.4 presents the survival function of prices, that is, the proportion of goods that remain unchanged for t periods or more. Figure 4.4(a) shows that the function declines rapidly, meaning that most prices have a short duration. After the first month, more than 25 % of goods have changed their prices. By the second month, more than half of the goods have done so, and by the fourth month, only 25 % of goods remain unchanged.

When analyzing by chain, Figure 4.4(b) shows that the proportion of goods in the Frigo chain changed by 25 % in the first month, while El Clon modified almost half of its goods' prices in the same period. This analysis reveals that prices in the Frigo chain are more rigid than those in other chains, highlighting the existence of heterogeneity among chains.

Figura 4.4: Survival Curve Analysis

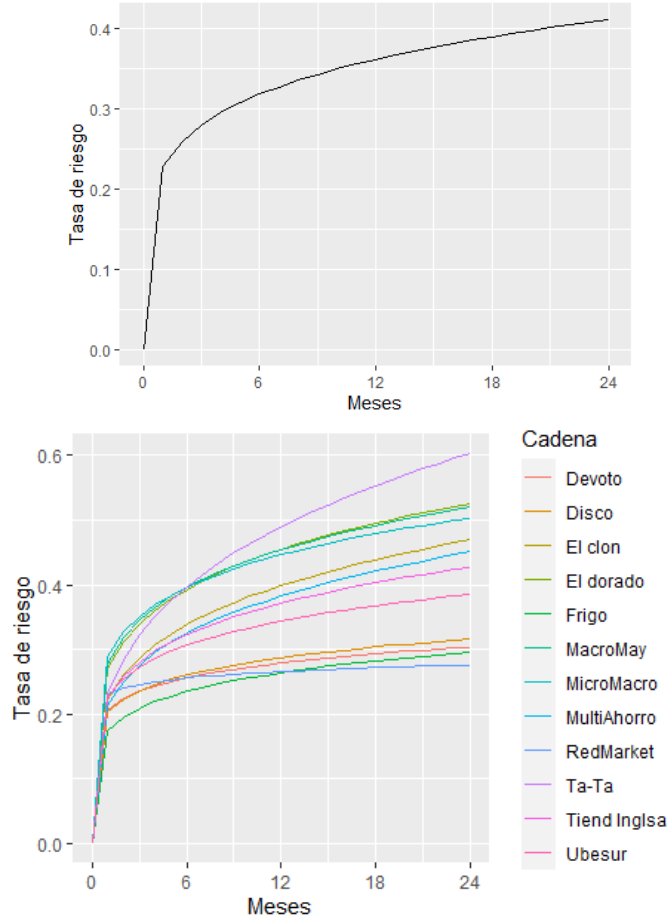


(a) Survival Curve (b) Survival Curve by Chain

Figure 4.5 presents the estimated Hazard function, which returns the probability of a price change given that the price remained unchanged for t periods.

Results show that the Hazard function has a positive slope (Figure 4.5), which is consistent with SDP models and aligns with the results of Cavallo (2009) and Borraz et al. (2012). However, these findings differ from those of Nakamura and Steinsson (2008), who reports negative slopes. This discrepancy can be explained by the fact that this study was conducted in a moderately inflationary economy, whereas Nakamura and Steinsson analyzed a low-inflation context. Cavallo argues that permanent shocks in moderately inflationary economies can cause deviations from the optimal price, and these tend to accumulate over time.

Figura 4.5: Hazard Function



(a) Hazard Rate (b) Hazard Rate by Chain

4.2.3. Synchronization of Price Changes

Table 4.6 presents the results of the synchronization indicator for chains, following the strategy described in Section 3.1.3.

There is heterogeneity in the level of synchronization among chains. El Clon exhibits the highest levels of price synchronization, followed by Disco and Devoto. The chains with the lowest synchronization levels are Ubesur and Red Market.

This pattern does not appear to be explained by the number of stores, suggesting that it may be related to the organizational structure of the chains. Ubesur is a group of supermarkets, whereas Devoto is a centralized chain.

Tabla 4.6: Price Synchronization Among Stores Belonging to the Same Chain

Chain	Stores	Average FK
Devoto	24	0.71
Disco	28	0.73
El Clon	14	0.75
El Dorado	44	0.60
Frigo	7	0.65
Macromercado Mayorista	7	0.66
Micro Macro	10	0.60
Multi Ahorro	48	0.67
Red Market	12	0.54
Ta - Ta	43	0.61
Tienda Inglesa	10	0.69
Ubesur	20	0.52

Source: Author's calculations based on DGC data.

4.2.4. Price Positioning

Another interesting aspect of the pricing strategy is related to the ability of retail outlets to set prices above or below the market price. Table 4.7 shows the weighted average price positioning by chain. It can be observed that the chain Tienda Inglesa has prices above the market average, while El Clon is positioned below the market average.

Tabla 4.7: Price Positioning

Chain	Price Positioning	height
0.99 Devoto		
0.99 Disco		
0.93 El Clon		
0.93 El Dorado		
1.00 Frigo		
0.97 Macromercado Mayorista		
0.99 Micro Macro		
1.00 Multi Ahorro		
1.00 Red Market		
0.96 Ta - Ta		
1.00 Tienda Inglesa		
0.99 Ubesur		

Source:

calculated by the author based on DGC data.

Capítulo 5

Final Considerations

This study aimed to contribute to the body of research analyzing supermarket chains' pricing strategies in Uruguay, using a sample of 154 products across 267 supermarkets nationwide from January 2011 to December 2019.

The results show that, on average, chains change 26 % of their prices each month, which corresponds to a duration of 3.99 months, and the average magnitude of these changes is 3.89 %. Although this duration is slightly higher than the 2.5 months found by [Borraz et al. \(2012\)](#), the discrepancy may be explained by the fact that their study used daily data, whereas this work used the monthly mode of prices. Prices in Uruguay are less frequently adjusted compared to the findings of [Chaumont et al. \(2011\)](#), where prices have an average duration of 1.95 months. Comparing with [Cavallo \(2018\)](#), which uses data obtained through web scraping, the monthly duration in the United States is 4.7 months, Argentina 3.43, Brazil 2.03, Chile 4.38, and Colombia 2.29.

When analyzing price changes by sign, positive changes are observed to be more frequent than negative ones. The analysis by the supermarket chain reveals an inverse relationship between the frequency of price changes and their magnitude: chains that change prices less frequently tend to implement changes of greater magnitude. The chain Micro Macro makes changes more frequently and of smaller size, whereas Frigo exhibits the opposite pattern.

These strategies appear to be determined at the chain level, as the detailed analysis of individual stores shows that they tend to cluster by chain in terms of frequency and magnitude, with a low level of overlap. The calculation of standardized distance allowed us to identify chains that are more likely to have homogeneous strategies among their stores. It is observed that

supermarkets belonging to El Clon exhibit less variation in their price-change strategies—both increases and decreases—whereas those in Red Market show greater disparity in price increases.

The survival and hazard analysis suggests that strategies may be aligned with the STD model. Duration and survival curves revealed the presence of chains with greater price rigidity.

The positive slope of the hazard function indicates that chains do not follow a TDP strategy; these results are consistent with the findings of [Borraz et al. \(2012\)](#).

Regarding price synchronization, heterogeneity was observed among chains. The FK index ranges between 0.52 and 0.75, indicating that no chain exhibits perfect synchronization. These results are lower than those reported by [Borraz et al. \(2012\)](#) for Uruguay, who found FK indices ranging from 0.84 to 0.96. This study also contrasts with [Chaumont et al. \(2011\)](#) for Chile, where synchronization ranged from 0.36 to 0.96.

Finally, through price positioning analysis, chains were identified that set prices above the market average. Tienda Inglesa, on average, exhibits the highest prices in the market.

Although this study provided updated measurements on pricing and synchronization strategies, examining differences across chains, it is essential to acknowledge some limitations.

While the analyzed goods followed a classification that allowed for some degree of homogeneity, the composition of the product basket offered by different chains may vary, targeting different consumer segments. It was not possible to incorporate the demand faced by each store, which could influence pricing decisions.

Additionally, the reasons behind the different strategies identified were not addressed. Future research could explore these dimensions by incorporating, for example, the level of competition among retail stores using geo-referenced information associated with store locations.

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ANEXOS

Listados de productos

Tabla 1: List of Products

Product	Item	
Food	Sunflower Oil	Sunflower Oil – Óptimo
Food	Sunflower Oil	Sunflower Oil – Río de la Plata
Food	Sunflower Oil	Sunflower Oil – Uruguay
Food	Corn Oil	Corn Oil – Delicia
Food	Corn Oil	Corn Oil – Río de la Plata
Food	Corn Oil	Corn Oil – Salad
Food	Soybean Oil	Soybean Oil – Condesa
Food	Soybean Oil	Soybean Oil – Río de la Plata
Food	Soybean Oil	Soybean Oil – Salad
Food	Rice	White Rice – Aruba
Food	Rice	White Rice – Blue Patna
Food	Rice	White Rice – Green Chef
Food	Rice	White Rice – Pony
Food	Rice	White Rice – Saman Blanco
Food	Rice	White Rice – Vidarroz
Food	Peas	Canned Peas – Campero
Food	Peas	Canned Peas – Cololó
Food	Peas	Canned Peas – Nidemar
Food	Sugar	White Sugar – Azucarlito
Food	Sugar	White Sugar – Bella Unión
Food	Coffee	Packaged Coffee – Águila
Food	Coffee	Packaged Coffee – Chaná
Food	Coffee	Packaged Coffee – Saint
Food	Meat	Beef Chuck – With Bone
Food	Meat	Beef Chuck – Boneless

Food	Meat	Ground Beef – Up to 5 % Fat
Food	Meat	Ground Beef – Up to 20 % Fat
Food	Meat	Beef Round – With Bone – No Brand
Food	Meat	Beef Round – Boneless – No Brand
Food	Meat	Beef Shoulder – With Bone – No Brand
Food	Meat	Beef Shoulder – Boneless – No Brand
Food	Meat	Beef Eye of Round – No Brand
Food	Meat	Beef Shank – With Bone – No Brand
Food	Sausages	Sausages – Cattivelli
Food	Sausages	Sausages – Centenario
Food	Sausages	Sausages – La Familia (Yellow Thread)
Food	Cocoa	Cocoa – Copacabana
Food	Cocoa	Cocoa – Vascolet
Food	Dulce de Leche	Packaged Dulce de Leche – Conaprole
Food	Dulce de Leche	Packaged Dulce de Leche – Los Nietitos
Food	Dulce de Leche	Packaged Dulce de Leche – Manjar
Food	Quince Paste	Packaged Quince Paste – Los Nietitos
Food	Egg Noodles	Dry Egg Noodles – Adria
Food	Egg Noodles	Dry Egg Noodles – Cololó
Food	Egg Noodles	Dry Egg Noodles – Las Acacias
Food	Semolina Noodles	Dry Semolina Noodles – Adria
Food	Semolina Noodles	Dry Semolina Noodles – Las Acacias
Food	Semolina Noodles	Dry Semolina Noodles – Puritas
Food	Frankfurters	Short Vacuum-Packed Frankfurters –
Food	Frankfurters	Short Vacuum-Packed Frankfurters –
Food	Frankfurters	Short Vacuum-Packed Frankfurters –
Food	Water Crackers	Water Crackers – Famosa
Food	Water Crackers	Water Crackers – Maestro Cubano
Food	Hamburgers	Beef Hamburgers – Burgy
Food	Hamburgers	Beef Hamburgers – Paty
Food	Hamburgers	Beef Hamburgers – Schneck
Food	Corn Flour	Corn Flour – Gourmet
Food	Corn Flour	Corn Flour – Presto Pronta Arcor
Food	Corn Flour	Corn Flour – Puritas
Food	Wheat Flour (Type 000)	Wheat Flour – Cañuelas
Food	Wheat Flour (Type 000)	Wheat Flour – Cololó

Food	Wheat Flour (Type 0000)	Wheat Flour – Cañuelas
Food	Wheat Flour (Type 0000)	Common Wheat Flour – Cololó
Food	Wheat Flour (Type 0000)	Common Wheat Flour – Primor
Food	Ice Cream	Family Ice Cream – Conaprole
Food	Ice Cream	Family Ice Cream – Crufi
Food	Ice Cream	Family Ice Cream – Gebetto
Food	Eggs	Brown Eggs – El Jefe
Food	Eggs	Brown Eggs – Prodhin
Food	Eggs	Brown Eggs – Super Huevo
Food	Ham	Cooked Ham (Non-Artisanal) – Cati
Food	Ham	Cooked Ham (Non-Artisanal) – Ottonello
Food	Leonesa Sausage	Leonesa Sausage – La Constancia
Food	Leonesa Sausage	Leonesa Sausage – Ottonello
Food	Leonesa Sausage	Leonesa Sausage – Schneck
Food	Butter	Butter – Calcar
Food	Butter	Unsalted Butter – Conaprole
Food	Butter	Butter – Kasdorf
Food	Margarine	Margarine – Doriana Nueva
Food	Margarine	Margarine – Flor
Food	Margarine	Margarine – Primor
Food	Mayonnaise	Regular Mayonnaise – Fanacoa
Food	Mayonnaise	Regular Mayonnaise – Hellmann's
Food	Mayonnaise	Regular Mayonnaise – Uruguay
Food	Jam	Peach Jam – Dulciora
Food	Jam	Peach Jam – El Hogar
Food	Jam	Peach Jam – Los Nietitos
Food	Bread	Baguette Bread – No Brand
Food	Sliced Bread	Sliced Bread – Los Sorchantes
Food	Sliced Bread	White Milk Bread – Bimbo
Food	Sliced Bread	White Milk Bread – Pan Catalán
Food	Fish	Fresh Fish – Hake Fillets – No Brand
Food	Chicken	Whole Chicken with Giblets – Avícola
Food	Chicken	Whole Chicken with Giblets – Tres A
Food	Chicken	Whole Chicken – Tenent
Food	Tomato Puree	Tomato Puree – Conaprole
Food	Tomato Puree	Concentrated Tomato Puree – De Ley

Food	Tomato Puree	Tomato Puree – Gourmet
Food	Grated Cheese	Grated Cheese – Artesano
Food	Grated Cheese	Grated Cheese – Conaprole
Food	Grated Cheese	Grated Cheese – Milky
Food	Salt	Fine Iodized and Fluoridated Salt – S
Food	Salt	Fine Iodized and Fluoridated Salt – T
Food	Salt	Fine Iodized and Fluoridated Salt – U
Food	Tea	Black Tea Bags – Hornimans
Food	Tea	Black Tea Bags – La Virginia
Food	Tea	Black Tea Bags – President
Food	Yerba Mate	Regular Yerba Mate – Baldo
Food	Yerba Mate	Regular Yerba Mate – Canarias
Food	Yerba Mate	Regular Yerba Mate – Del Cebador
Food	Yogurt	Yogurt – Conaprole BIO TOP
Food	Yogurt	Semi-Skimmed Peach Yogurt – Calca
Food	Yogurt	Semi-Skimmed Peach Yogurt – Parma
Alcoholic Beverages	Beer	Beer – Patricia
Alcoholic Beverages	Beer	Beer – Pilsen
Alcoholic Beverages	Beer	Beer – Zillertal
Alcoholic Beverages	Wine	Common Red Wine (Tetra Pak) – Fa
Alcoholic Beverages	Wine	Common Red Wine (Tetra Pak) – Sa
Alcoholic Beverages	Wine	Common Red Wine (Tetra Pak) – Ta
Non-Alcoholic Beverages	Water	Table Water – Sparkling – Salus (2 L
Non-Alcoholic Beverages	Water	Table Water – Sparkling – Matutina
Non-Alcoholic Beverages	Water	Table Water – Sparkling – Nativa (2
Non-Alcoholic Beverages	Soft Drink	Soda – Coca-Cola (1.5 L)
Non-Alcoholic Beverages	Soft Drink	Soda – Coca-Cola (2.25 L)
Non-Alcoholic Beverages	Soft Drink	Soda – Nix (1.5 L)
Non-Alcoholic Beverages	Soft Drink	Soda – Pepsi (1.5 L)
Non-Alcoholic Beverages	Soft Drink	Soda – Pepsi (2 L)
Cleaning	Shampoo	Shampoo – Fructis
Cleaning	Shampoo	Shampoo – Sedal
Cleaning	Shampoo	Shampoo – Suave
Cleaning	Deodorant	Aerosol Deodorant – Axe Musk
Cleaning	Deodorant	Aerosol Deodorant – Dove Original
Cleaning	Deodorant	Aerosol Deodorant – Rexona Active

Cleaning	Dishwashing Detergent	Dishwashing Detergent – Deterjane L
Cleaning	Dishwashing Detergent	Dishwashing Detergent – Hurra Never
Cleaning	Dishwashing Detergent	Dishwashing Detergent – Protergente
Cleaning	Bleach	Sodium Hypochlorite – Agua Jane
Cleaning	Bleach	Sodium Hypochlorite – Sello Rojo
Cleaning	Bleach	Sodium Hypochlorite – Solución Crist
Cleaning	Laundry Soap	Laundry Bar Soap – Bull Dog
Cleaning	Laundry Soap	Laundry Bar Soap – Nevex
Cleaning	Laundry Soap	Laundry Bar Soap – Primor
Cleaning	Toilet Soap	Toilet Soap – Astral Plata
Cleaning	Toilet Soap	Toilet Soap – Palmolive
Cleaning	Toilet Soap	Toilet Soap – Rexona
Cleaning	Powder Detergent	Machine Washing Powder – Drive
Cleaning	Powder Detergent	Machine Washing Powder – Nevex
Cleaning	Powder Detergent	Machine Washing Powder – Skip
Cleaning	Toilet Paper	Toilet Paper – Single Ply – Elite
Cleaning	Toilet Paper	Toilet Paper – Single Ply – Higienol
Cleaning	Toilet Paper	Toilet Paper – Single Ply – Sin Fin
Cleaning	Toothpaste	Toothpaste – Colgate Herbal Whiteni
Cleaning	Toothpaste	Toothpaste – Kolynos Triple Action
Cleaning	Toothpaste	Toothpaste – Pico Jenner Plus

Fuente: calculado por la autora en base a datos de la DGC.

Results

Price Variation

Table 2: Magnitude of Price Changes by Category

Category	Mean	SD	Min	Max	Kurtosis	Skewness
Food	0.79	6.46	-67.58	306.78	52.30	2.90
Alcoholic Beverages	0.78	4.46	-42.35	73.47	41.12	1.84
Non-Alcoholic Beverages	0.72	3.66	-54.55	72.73	53.99	0.97
Cleaning Products	1.02	8.94	-68.97	249.10	30.09	2.22

Source: Author's calculations based on DGC data.

Table 3: Magnitude of Price Changes by Product

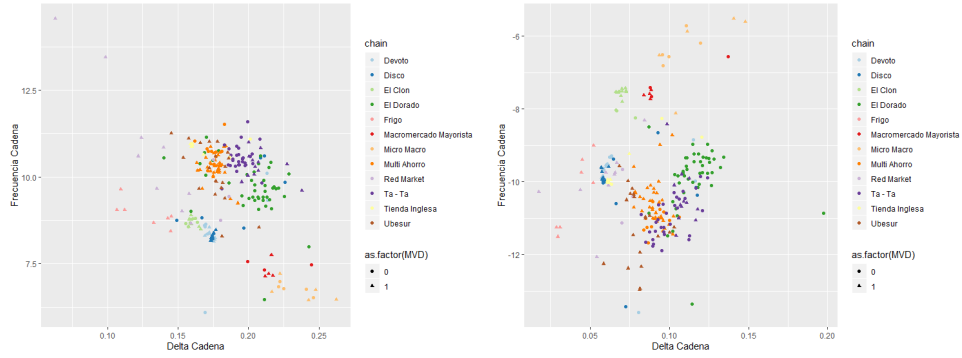
Product	Mean	SD	Min	Max	Kurtosis	Skewness
Sunflower Oil	0.33	4.35	-40.96	46.88	15.29	0.46
Corn Oil	0.61	7.40	-49.28	97.14	56.63	3.51
Soy Oil	0.26	4.74	-36.51	48.00	15.71	1.04
Water	0.66	3.73	-54.55	44.00	56.82	-2.01
Rice	0.72	8.54	-50.00	89.47	18.63	1.66
Peas	1.23	12.76	-54.55	130.00	33.36	3.44
Sugar	0.22	3.14	-44.44	56.10	37.91	-1.52
Coffee	0.60	3.99	-55.05	142.86	66.53	0.64
Meat	0.89	6.51	-67.58	129.41	25.16	1.69
Beer	0.90	3.53	-26.13	28.30	13.31	0.96
Shampoo	1.10	9.90	-66.63	249.10	42.74	2.93
Sausages	0.77	4.94	-44.85	85.54	17.65	1.37
Cocoa	0.80	6.05	-38.12	63.53	12.24	1.32
Deodorant	1.18	10.72	-34.17	70.89	4.26	0.54
Detergent	0.98	7.86	-44.44	97.44	9.18	0.95
Dulce de Leche	0.65	5.58	-65.37	188.77	30.35	1.37
Quince Paste	0.86	6.34	-26.92	40.62	8.88	0.79
Egg Noodles	0.70	5.73	-42.31	65.83	12.91	0.91

Semolina Noodles	0.94	6.57	-50.13	100.50	13.03	1.33
Frankfurters	0.98	5.76	-66.51	306.78	471.50	13.53
Water Crackers	1.21	7.49	-42.86	116.67	61.37	5.47
Hamburgers	0.69	5.13	-65.70	116.67	44.87	3.05
Corn Flour	1.08	10.38	-57.50	135.29	33.39	3.26
Wheat Flour 000	0.88	7.28	-36.67	153.33	75.29	4.05
Wheat Flour 0000	0.91	7.98	-42.86	110.00	10.42	1.05
Ice Cream	0.85	7.01	-46.86	88.17	13.52	1.11
Bleach	0.83	5.25	-46.30	118.75	30.95	1.98
Eggs	0.97	5.79	-33.96	49.17	10.55	1.09
Laundry Soap	1.09	6.90	-41.18	146.42	152.42	7.77
Toilet Soap	1.02	8.16	-68.97	100.00	20.47	1.28
Powdered Detergent	0.82	6.76	-42.86	78.94	11.80	1.10
Ham	0.72	7.04	-63.01	144.44	25.71	0.21
Leonesa (Pasta)	0.84	4.57	-40.84	46.01	15.47	1.15
Butter	0.66	5.27	-42.86	96.97	23.11	1.82
Margarine	1.03	5.80	-36.36	138.76	34.83	2.62
Mayonnaise	0.85	7.04	-42.86	58.93	12.03	1.22
Jam	0.88	7.62	-63.46	173.68	80.58	4.97
Bread	0.86	5.59	-48.86	137.41	79.68	4.08
Sliced Bread	0.98	5.14	-42.16	145.71	152.91	7.18
Toilet Paper	0.92	6.04	-48.98	84.00	23.61	1.88
Toothpaste	1.23	13.82	-59.92	164.50	19.83	2.19
Fish	0.89	4.93	-32.26	58.20	24.45	2.28
Chicken	0.98	8.20	-36.70	72.46	7.06	0.68
Tomato Paste	0.62	6.45	-54.39	103.85	14.51	1.05
Grated Cheese	0.66	3.51	-38.78	46.67	34.11	2.30
Soft Drink	0.76	3.61	-33.67	72.73	51.73	2.96
Salt	0.89	3.67	-51.67	87.50	37.59	3.19
Tea	0.82	5.06	-59.18	110.00	37.96	1.40
Wine	0.64	5.29	-42.35	73.47	40.83	2.05
Yerba Mate	0.92	3.52	-36.80	56.57	24.45	2.13
Yogurt	0.87	8.42	-41.09	69.75	10.95	1.16

Source: Author's calculations based on DGC data.

Delta and Frequency by Supermarket

Figura 1: Average Magnitude and Frequency of Price Changes



(a) Monthly Positive Changes (b) Monthly Negative Changes

Price Duration

Tabla 4: Price Duration by Category

Category	Average
Food	2.48
Alcoholic Beverages	2.37
Non-Alcoholic Beverages	2.65
Cleaning Products	1.67

Source: Author's calculations based on DGC data.

Tabla 5: Price Duration by Product

Product	Average	Product	Average
Sunflower Oil	2.01	Ice Cream	1.62
Corn Oil	1.92	Bleach	2.71
Soy Oil	2.10	Eggs	2.78
Water	3.26	Laundry Soap	2.48
Rice	1.97	Toilet Soap	2.11
Peas	1.94	Powdered Detergent	1.61
Sugar	6.93	Ham	2.11
Coffee	5.00	Leonesa (Pasta)	3.12
Meat	1.69	Butter	2.33
Beer	2.21	Margarine	2.28
Shampoo	1.68	Mayonnaise	1.56
Sausages	2.98	Jam	2.74
Cocoa	2.81	Bread	4.18
Deodorant	0.98	Sliced Bread	2.97
Detergent	1.37	Toilet Paper	2.17
Dulce de Leche	2.19	Toothpaste	1.40
Quince Paste	2.19	Fish	4.09
Egg Noodles	2.00	Chicken	1.02
Semolina Noodles	2.65	Tomato Paste	2.21
Frankfurters	3.38	Grated Cheese	3.88
Water Crackers	4.29	Soft Drink	2.37
Hamburgers	4.36	Salt	5.40
Corn Flour	2.73	Tea	4.69
Wheat Flour 000	2.39	Wine	2.59
Wheat Flour 0000	1.63	Yerba Mate	3.85
⋮		Yogurt	1.63

Source: Author's calculations based on DGC data.

Anexo 1

R Code

The script used, as well as the programmed functions, are available on the author's
GitHub repository.